Quadrilaterals

The big idea

A **quadrilateral** has four sides and four angles that add up to **360°**. Special quadrilaterals (parallelogram, rectangle, rhombus, square) have extra properties, and the **Mid-point Theorem** links the midpoints of a triangle's sides.

The parallelogram and its properties

IN A PARALLELOGRAM

A diagonal divides it into two **congruent triangles**.
Opposite sides are equal, and **opposite angles are equal**.
The **diagonals bisect each other**.
Each converse is also true; for example, if a quadrilateral's diagonals bisect each other, it is a parallelogram. So is a quadrilateral with **one pair of opposite sides equal and parallel**.

Shape	Extra properties
Rectangle	All angles 90°; diagonals equal
Rhombus	All sides equal; diagonals perpendicular
Square	All sides equal, all angles 90°; diagonals equal and perpendicular

ONE ANGLE OF A PARALLELOGRAM IS 70°. FIND THE OTHERS.

The opposite angle is 70°. Adjacent angles are supplementary: 110°, 110°.
Angles: **70°, 110°, 70°, 110°**.

The Mid-point Theorem

MID-POINT THEOREM

The line segment joining the **midpoints of two sides** of a triangle is **parallel to the third side** and **half of it**.
Converse: a line through the midpoint of one side, parallel to another side, bisects the third side.

IN △ABC, D AND E ARE MIDPOINTS OF AB AND AC, AND BC = 10 CM. FIND DE.

DE ∥ BC and DE = ½ BC = **5 cm**.

THE TRIANGLE OF MIDPOINTS

Joining the midpoints of the three sides of a triangle gives four congruent triangles, each similar to the original.

Quick recap

- Angle sum of a quadrilateral is 360°.
- Parallelogram: opposite sides and angles equal; diagonals bisect each other.
- Mid-point Theorem: segment joining midpoints ∥ third side and half of it.

Practice questions

1. The sum of the angles of a quadrilateral is
   (a) 180° (b) 360° (c) 540° (d) 90°
2. The diagonals of a rhombus
   (a) are equal (b) bisect each other at right angles (c) are parallel (d) do not bisect each other
3. Adjacent angles of a parallelogram are
   (a) equal (b) supplementary (c) complementary (d) always 90°
4. D and E are midpoints of AB and AC in $\triangle ABC$, and DE = 4 cm. Then BC is
   (a) 4 cm (b) 2 cm (c) 8 cm (d) 16 cm
5. A quadrilateral whose diagonals bisect each other is a
   (a) parallelogram (b) trapezium (c) kite (d) cyclic quadrilateral only

Answers: 1. b 2. b 3. b 4. c 5. a